Ambient Systems

Business Plan Technology Slides

Motivation

- Determining trade-off between objectives is difficult:
  - Energy Use
  - Indoor Air Quality (IAQ)
  - Thermal Comfort
- Multi-zone Variable Air Volume (VAV) air-conditioning systems have a large number of set-points in different control loops
- Recommended setpoints by experienced building operators or guidelines cannot guarantee optimal energy use, IAQ, and ideal crop environment
- Computing challenge:
  - Computational load as number of variables increases
  - Global optimum is hard to find due to non-convexity
  - Existing BAS are centralized, which are limited in terms of scalability
- Two variables are most important for optimization:
  - Fresh air ratio of supply air
  - Temperature set-points of individual zones
- These can be set in multiple ways:
  - Based on experience by building operators
  - Technical specifications and guidance

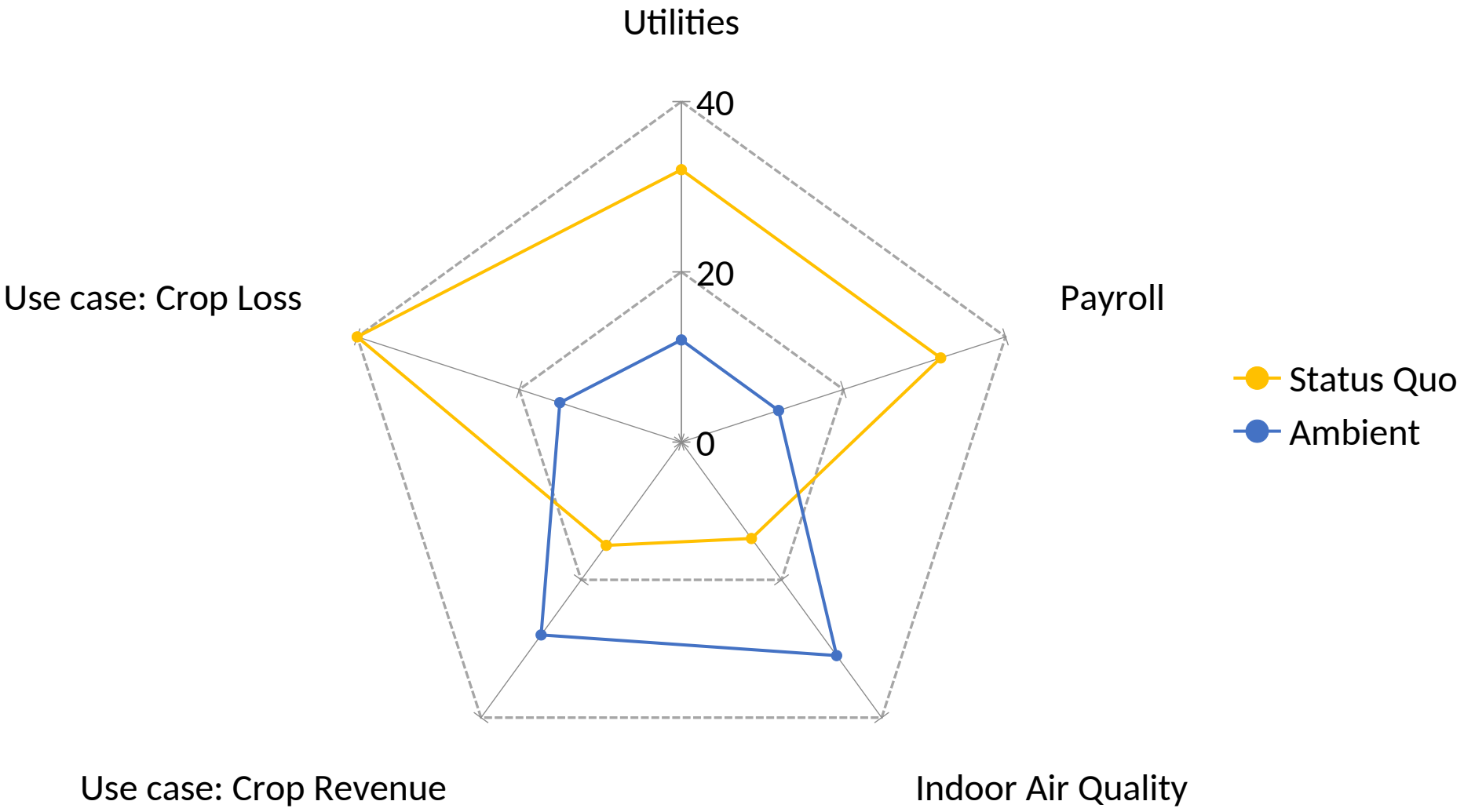
Solution

- Propose: multi-agent distributed optimization method
- Three novel schemes:
  - Temperature set-point reset scheme adopts a linear rule to correlate set-points in individual zones
  - Multi-objective optimization scheme optimizes the fresh air ratio of supply air and temperature
  - Multi-agent distributed optimization scheme solves the optimization problem in a distributed manner

    :
  - Particle Swarm Algorithm (PSA) using quad-objective optimization among energy use, indoor air temperature, relative humidity, and CO2 concentration level → output (1) set-points of supply air temperature and (2) static pressure of an Air Handling Unit (AHU)
  - Sequential Methods that uses energy use, indoor ozone, PM2.5 and HCHO → output (1) set-point of air exchange rate
  - Sparse (sFA) and sparse Augmented (sAFA)

Methods

- Degrees of freedom:
  - Location: Zone (3D Matrix)
  - Time: Sampling Rate


Utilities
40
20
0
Use case: Crop Loss
Payroll
Status Quo
Ambient
Use case: Crop Revenue
Indoor Air Quality

Optimization Function

- Optimization Objectives (O):
  - Utilities: Measured by Energy Consumption ($)
  - Indoor Air Quality: Measured by CO2
  - Productivity: Measured by Crop Values ($)

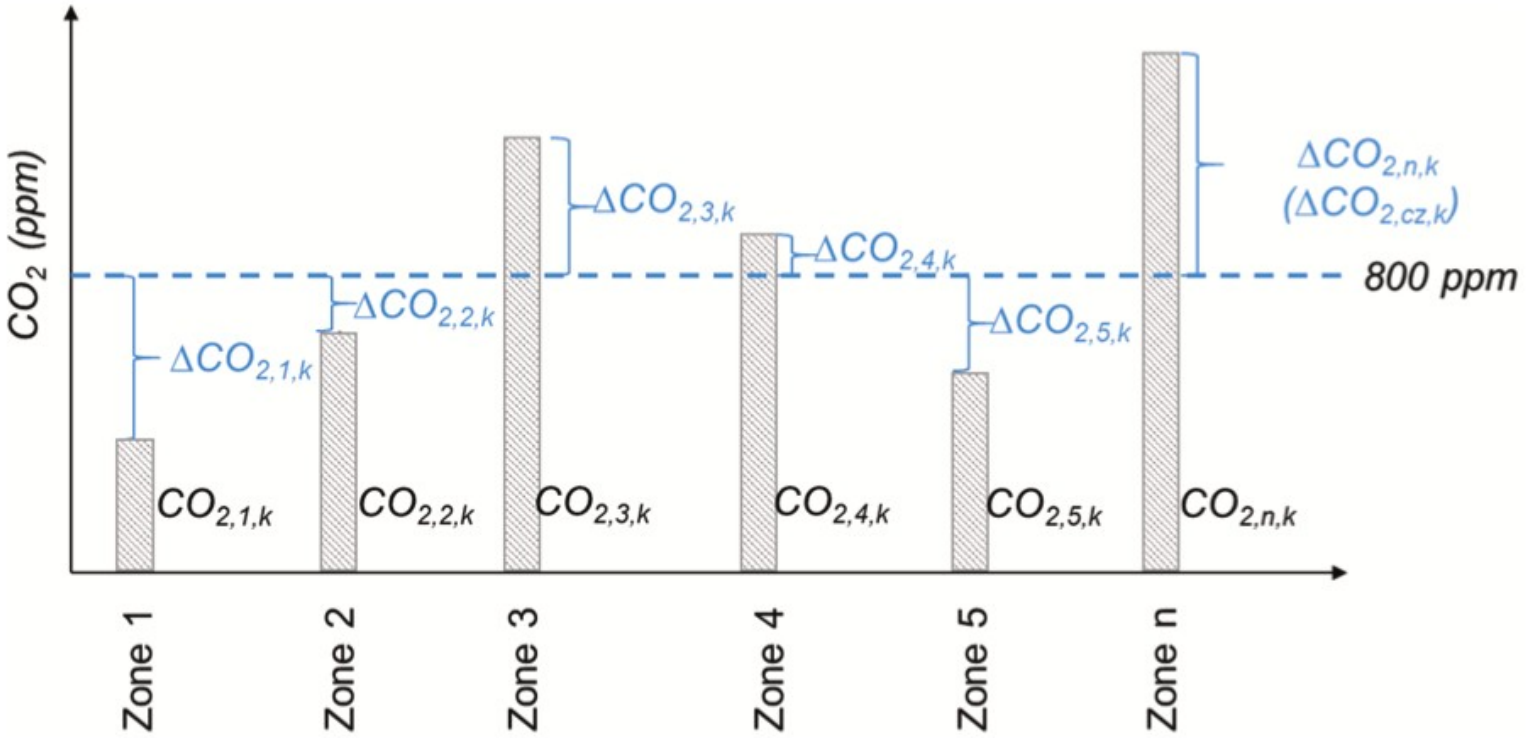
$$Objective = \alpha_{Utilities} * O_{Utilities} + \alpha_{IAQ} * O_{IAQ} + \alpha_{Productivity} * O_{Productivity}$$

Utilities (Electricity)

- Sample equipment (i):
  - Chiller Plant: provides cooling to all the zones
  - AHU Fan: Delivering Airflow to the zones
  - Auxiliary Devices: Fans, AC
- Sum the energy consumption (V*A) across time series (t: 0 to time interval) and relevant equipment (j: 1 to # of equipment)

$$O_{Utilities} = \sum_{t=0}^{t} \sum_{i=1}^{i} V_i * A_i$$

IAQ



$$Obj_{IAQ} = \sum_{i=1}^{n} \max\{0, (CO2_i - TH_{CO2})\}$$

Productivity

- Need to work with AgTech experts to identify appropriate function for crop productivity
- Productivity of the farm can be calculated based on following variables:
  - Y: Crop Yield (kg / m2)
  - V: Crop Value ($ / kg)
- Summed across:
  - Crops (i): strawberries, lettuce
  - Harvest Tier (j): premium strawberries, standard strawberries, spoiled (e.g., due to mold or pests)
  - Time (t): harvest cycle (e.g., hours, days, weeks, months)

$$O_{Productivity} = \sum_{t=0}^{t} \sum_{i=1}^{i} \sum_{j=1}^{j} Y_{ij} * V_{ij}$$

Optimization Process

- Three steps:
  - Establish multi-agent system (define model)
  - Reformulated to multiple independent sub-problems
  - Iteration
- General: determine local optimal then global optimal
- Multiple control strategies:
  - #1: Fixed Setting
  - #2: User Adjusts
  - #3: Optimum Calculated by ML then Set
  - #4: Real-time calculations & adjustments

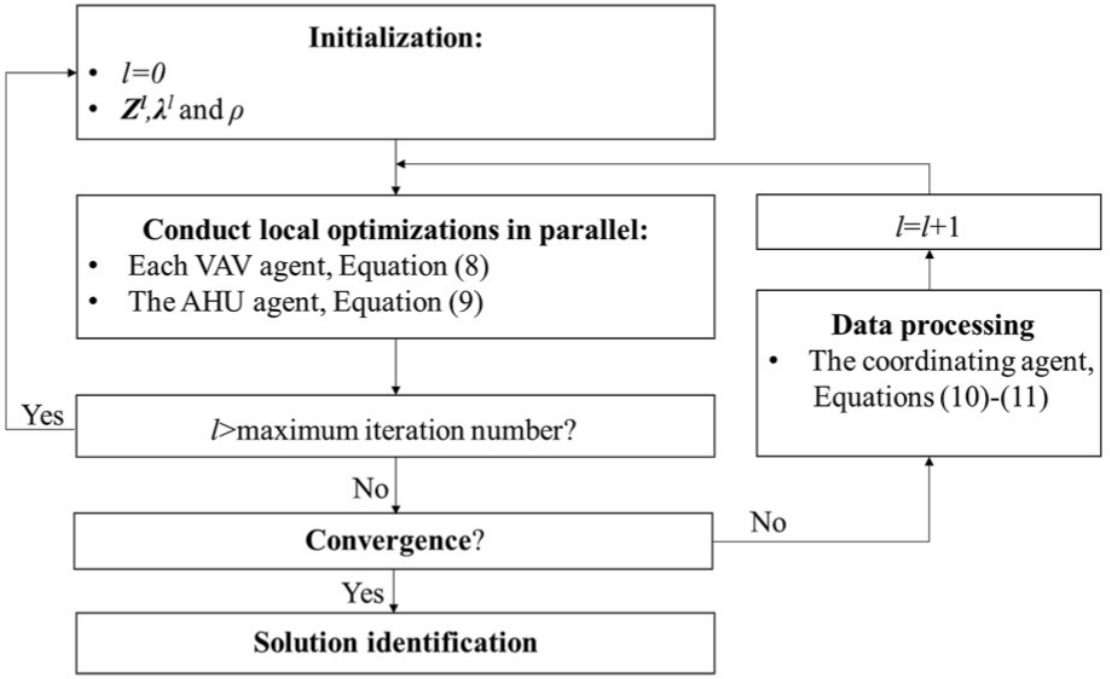
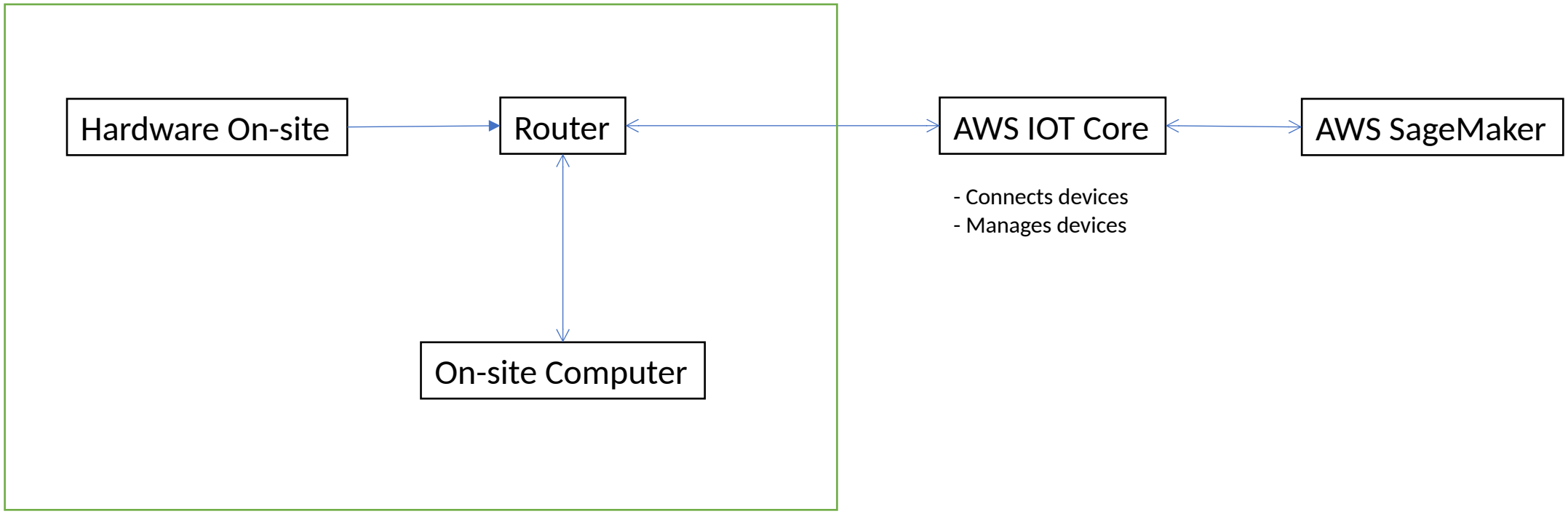


Fig. 6. Flowchart for the solution identification using ADMM.

MVP Architecture



Datasets

Equipment Data

Timestamp	Thermostat 1	Fan 1	Fan 2

Climate Data

Timestamp	Sensor 1	Sensor 2	Sensor 3

Crop Data

Timestamp	Number of Plants	Fruit per Plant	Fruit Quality Scale

Datasets: Simulation

Variable	Unit	Type	Distribution	Parameters
Equipment Setting				

Variable	**Unit**	**Type**	**Distribution**	**Parameters**
Inter-Arrival	sec	Continuous	$-0.001 + Exponential(\lambda)$	$\lambda = 1.35$
Base Station	units	Discrete	$DiscreteUniform(N)$	$N = 19$
Call Duration	sec	Continuous	$Exponential(\theta)$	$\theta = 119$
Location Between Base Stations	km	Continuous	$Uniform(\alpha, \beta)$	$\alpha = 0$ $\beta = 2$
Car Speed	km/hr	Continuous	$Triangular(a, b, c)$	$a = 70$ $b = 90$ $c = 110$

Table 1: Summary of Input Variable Distributions

Supervised Learning

Delivery Service Process

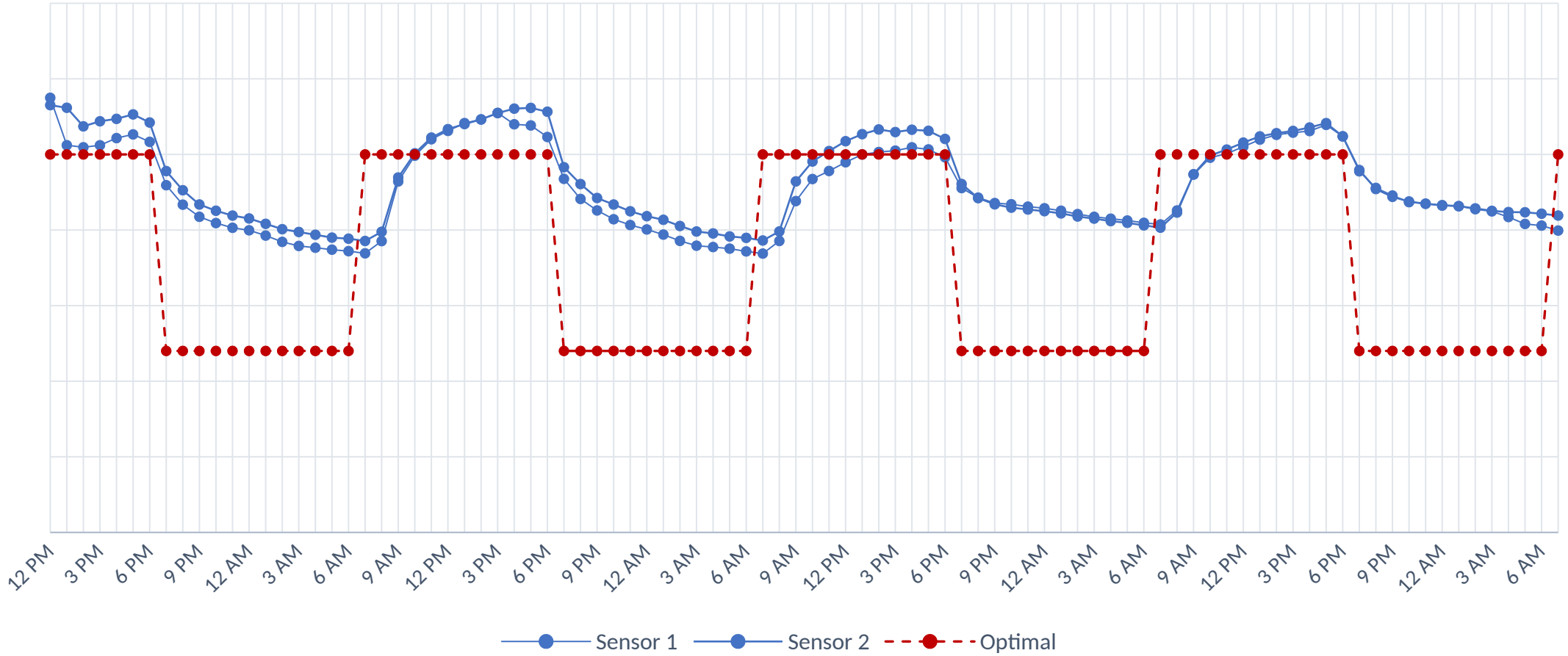
- Step 1: Requirements Gathering
- Step 2: IoT System Design (Consultation)
- Step 3: Vendor Management / Implementation
- Step 4: Software Subscription
- Step 5: Maintenance

Stream	V01 (Q1 2021)	V02 (Q2 2021)	V03 (Q3 2021)	Product Launch (Q4 2021)
Hardware	**Sensors** Collect essential parameters: Temperature, RH, CO2, and light intensity	**Sensors Extended** Collect additional parameters and increase number of sensors	**This is a sample text.** This is a sample text. Insert your description here.	Sensors Controls
Dashboard	**On-Site Monitoring** Displays sensor readings in real-time	**Online Monitoring** Real-time visualization, analytics, and notifications	**This is a sample text.** This is a sample text. Insert your description here.	Sample text
Machine Learning	**This is a sample text.** This is a sample text. Insert your description here.	**Deploy on the cloud** This is a sample text. Insert your description here.	**Trained Model** This is a sample text. Insert your description here.	Sample text

Appendix

Temperature

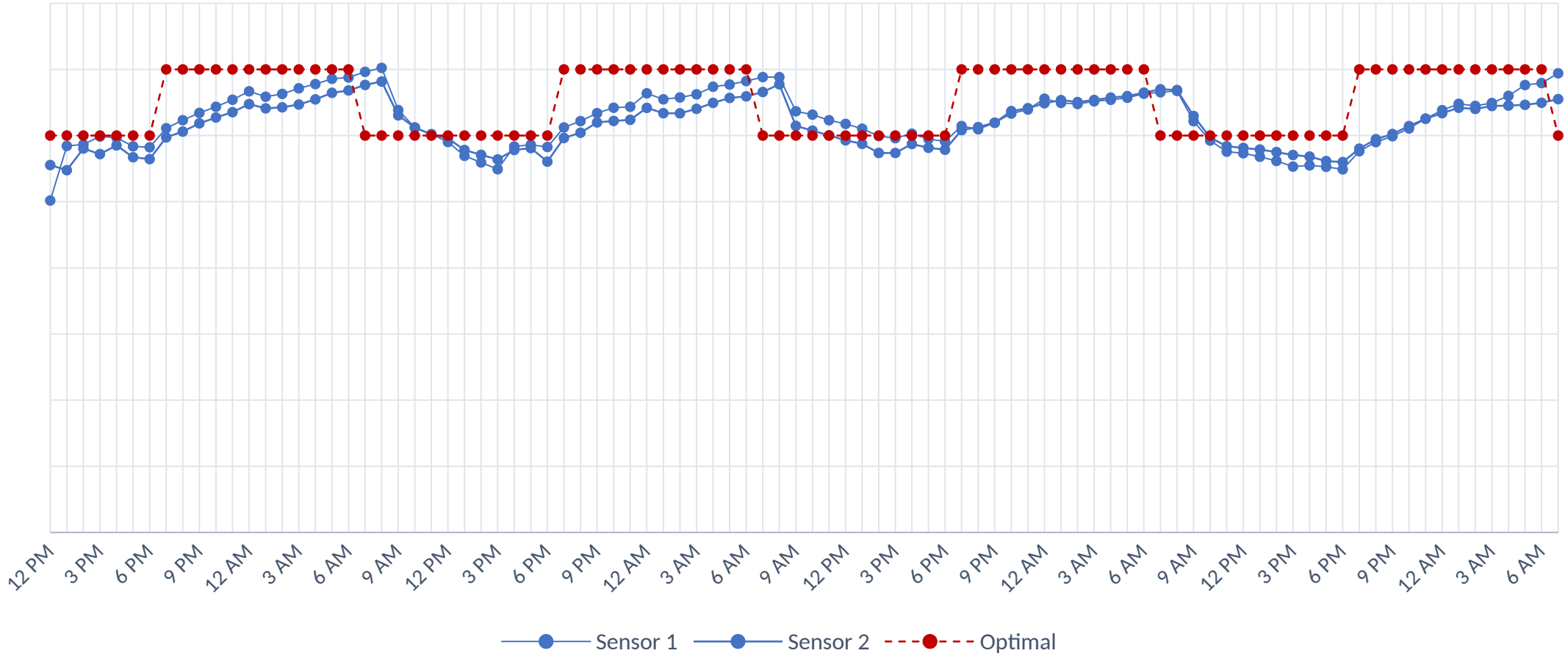
Leaf and petiole growth
Brighter leaf color
Greater sucrose
More biomass at the roots



Temperatures after bloom affect plant growth and fruit quality of strawberry, , 2000

Relative Humidity

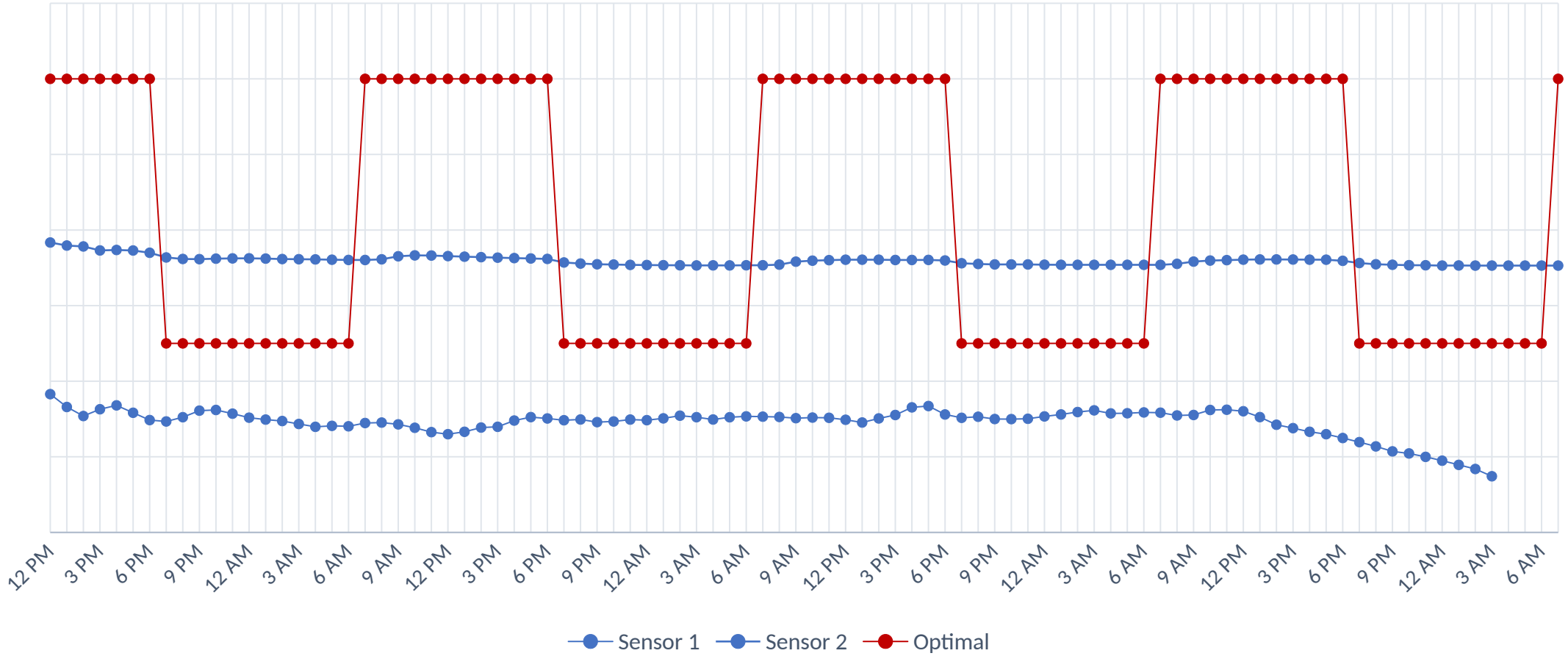
Avoid tip-burn and calyx burn
Avoid fungi
Avoid powdery mildew
Avoid leaf spot (bacterial blight)



Crop Guide: Growing Strawberry, , 2004

CO2

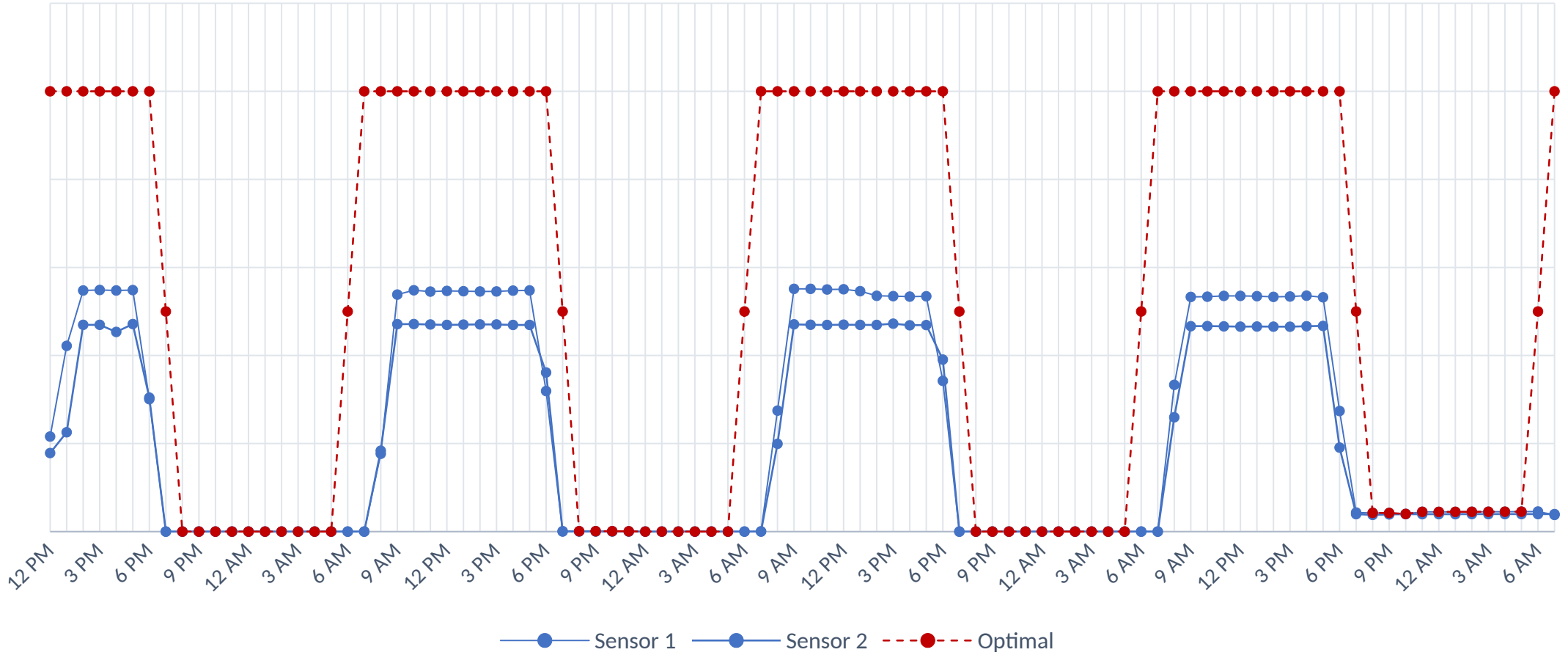
Increased photosynthesis
Earlier flowering
Higher fruit yields
Improved stem strength and flower size



Carbon Dioxide In Greenhouses, Ministry of Agriculture, Food and Rural Affairs, 2021
Carbon dioxide (CO2) applications in indoor growing,

Light Intensity

Increased photosynthesis



Effect Of Light Intensity, Temperature And Co2 Concentration On Photosynthesis And Growth Of In Vitro, International Society for Horticultural Science, 1989